

ZLIKHA, KazakhstanWMO: **359690**Lat: **45.222N**Lon: **66.877E**Elev: **138**StdP: **99.68**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.3	-22.4	-28.7	0.3	-24.9	-25.9	0.4	-21.5	17.8	-6.1	15.8	-12.4	2.2	50	0.559

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	38.6	19.3	37.1	18.8	35.5	18.4	20.9	34.2	20.1	33.4	19.4	32.7	3.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	12.4	24.7	15.5	11.2	24.6	14.0	10.1	24.6	61.0	34.0	58.1	33.5	55.6	32.6	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.9	10.2	8.8	-28.0	41.4	4.4	1.8	-31.2	42.7	-33.8	43.7	-36.2	44.7	-39.4	46.0		
(5)			-28.6	23.2	4.6	2.1	-31.9	24.8	-34.6	26.0	-37.2	27.2	-40.6	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-8.5	-6.1	3.6	12.9	19.5	24.8	26.4	25.1	18.0	9.7	1.1	-5.8
	DBStd	13.52	6.81	7.58	6.43	5.29	3.99	3.60	3.30	3.68	4.48	4.99	5.86	6.46
	HDD10.0	2099	574	452	211	32	1	0	0	0	2	68	270	489
	HDD18.3	3781	832	686	457	174	34	2	0	2	60	270	516	747
	CDD10.0	2153	0	0	13	118	295	445	509	468	242	58	4	0
	CDD18.3	792	0	0	0	10	70	197	251	213	49	1	0	0
	CDH23.3	11604	0	0	10	174	1054	2886	3730	2981	720	49	0	0
	CDH26.7	6251	0	0	2	48	432	1600	2182	1682	299	7	0	0
Wind	WSAvg	3.3	3.8	3.7	3.9	3.6	3.5	3.1	3.0	3.2	3.0	2.7	2.8	3.5
Precipitation	PrecAvg	251	31	30	29	26	22	16	12	6	5	16	27	31
	PrecMax	449	63	69	53	60	74	58	40	24	16	55	59	74
	PrecMin	136	9	6	6	5	5	3	1	0	1	2	6	9
	PrecStd	74	13	16	14	14	14	14	10	7	4	13	13	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	12.1	24.5	31.3	35.7	39.7	41.5	40.1	36.4	27.8	18.6	8.5
		MCWB	5.2	7.9	13.8	16.1	17.4	19.6	20.3	19.9	18.7	14.8	11.8	5.0
	2%	DB	3.2	7.8	19.1	27.8	33.0	37.8	39.0	37.8	32.6	24.5	14.8	5.3
		MCWB	1.8	4.8	10.9	14.4	16.9	19.0	19.4	19.0	16.7	12.8	9.5	3.2
	5%	DB	1.5	4.9	15.9	25.1	31.1	36.0	37.1	36.2	30.2	21.8	11.7	3.5
		MCWB	0.2	2.4	9.3	13.3	16.2	18.6	18.9	18.6	15.8	11.5	7.6	1.8
	10%	DB	0.1	2.6	13.0	22.6	29.1	34.2	35.4	34.5	27.6	18.9	9.0	1.6
		MCWB	-1.0	0.6	7.8	12.5	15.6	18.1	18.5	18.0	14.6	10.2	5.9	0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	8.4	13.9	17.7	20.1	22.1	21.8	22.0	19.2	15.2	13.0	5.7
		MCDB	7.4	11.9	22.7	26.7	29.2	33.6	34.8	34.1	34.2	25.5	16.5	7.6
	2%	WB	2.0	5.0	11.7	15.8	18.6	20.6	20.8	20.4	17.4	13.6	10.0	3.5
		MCDB	2.9	7.3	17.7	24.9	28.6	33.1	34.9	34.3	31.0	22.8	14.3	5.0
	5%	WB	0.4	2.7	10.0	14.4	17.4	19.7	20.1	19.5	16.1	12.1	7.8	1.9
		MCDB	1.2	4.7	14.9	22.9	28.2	32.7	33.6	33.8	28.9	20.3	11.2	3.3
	10%	WB	-1.0	0.8	8.2	13.1	16.5	18.9	19.3	18.6	14.9	10.6	6.2	0.3
		MCDB	0.1	2.3	12.5	21.2	27.1	31.9	32.7	33.0	26.5	18.0	8.9	1.4
Mean Daily Temperature Range	5% DB	MDBR	7.6	9.4	11.4	13.4	15.0	15.2	15.1	15.5	15.8	14.0	9.7	7.7
		MCDBR	6.6	10.6	15.8	17.0	17.9	17.6	17.1	17.7	18.8	17.9	13.6	8.7
		MCWBR	5.7	7.8	8.9	7.6	7.2	6.7	6.0	7.1	8.4	9.6	8.9	7.0
	5% WB	MCDBR	6.3	10.3	14.5	15.0	15.6	15.7	15.7	16.0	17.8	16.3	12.0	7.8
		MCWBR	5.6	8.0	8.9	7.3	7.3	6.9	6.2	7.1	8.5	9.3	8.2	6.5
Clear-Sky Solar Irradiance	taub	0.348	0.377	0.431	0.471	0.476	0.456	0.441	0.413	0.380	0.388	0.375	0.353	
	taud	2.064	2.025	1.984	1.923	1.955	2.062	2.128	2.187	2.230	2.128	2.115	2.090	
	Ebn at Noon	712	770	782	788	799	817	825	835	831	752	676	659	
	Edh at Noon	102	129	155	179	179	162	150	136	119	114	97	90	
All-Sky Solar Radiation	RadAvg	1.41	2.58	3.91	5.41	6.93	7.47	7.32	6.58	5.16	3.23	1.81	1.19	
	RadStd	0.30	0.32	0.31	0.40	0.38	0.36	0.33	0.24	0.18	0.20	0.18	0.18	